# MZUZU UNIVERSITY

## FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

### DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

### BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

## BICT 4701: SOFTWARE ENGINEERING

## SUPPLEMENTARY EXAMINATION

**DATE: 13 DECEMBER 2021** **TIME ALLOWED: 3 HOURS**

**INSTRUCTIONS**

i. There are 8 questions in this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Define the following terms:
    a. Software Process **[2 Marks]**
    b. Batch processing systems **[2 Marks]**
    c. Cloud computing **[2 Marks]**
    d. Use case diagrams **[2 Marks]**
    e. Sequence diagrams **[2 Marks]**

2. Discuss **two** advantages and **two** disadvantages of the following software process models:
    a. Waterfall model **[4 Marks]**
    b. Integration and configuration model **[4 Marks]**

3. Discuss **four** stages of Software Risk Management process and mention **one** major deliverable from each of the stages? (Use process diagram) **[10 Marks]**

4. Mphatso has a large DVD movie collection. Her friends like to borrow her DVD's, and she needs a way to keep track of who has what. She maintains a list of friends, identified by unique FID's (friend identifiers) and a list of DVD's, identified by DVDID's (DVD identifiers). With each friend is the name and the all-important telephone numbers which she can call to get the DVD back. With each DVD is the star actor name and title. Whenever a friend borrows a DVD, Mphatso will enter that fact into her database along with the date borrowed. Whenever the DVD gets returned, that fact, too, gets noted along with the date returned. Mphatso wants to keep a complete history of her friends' borrowing habits so that she can ask favors of the heavy borrowers (or perhaps refuse to make further loans to those who habitually don't return them quickly)
    a. Draw an E-R diagram for a database to help Mphatso with her business? **[15 Marks]**
    b. Which of the following questions can be answered from your ER diagram? Briefly justify the answers:
        i. Who are the heavy borrowers? **[2 Marks]**
        ii. Who habitually don't return DVD's quickly? **[2 Marks]**
        iii. Who borrowed a DVD multiple times? **[2 Marks]**
        iv. Who will be heavy borrowers? **[2 Marks]**
        v. Which DVDs are borrowed by someone at present? **[2 Marks]**

    c. List the **seven** major activities that must be done during the design phase **[7 Marks]**
    d. What are the **five** required skills of the systems analyst? **[5 Marks]**

5. Imagine that you have been given a contract as a software engineer to analyze the requirements of the Ticket Issuing System in the Library of Mzuzu University. Mention **three** functional requirements and **three** non-functional requirements of the ticket issuing system? **[6 Marks]**

6. Draw a sequence diagram modeling the interactions involved when a student register for a course at Mzuzu University. Assume that courses may have limited enrolment, so the registration process must include checks that places are available. Also the student needs to be authorized to accesses an online course catalog to find out about available courses. **[8 Marks]**

7. Consider your experience with a Bank ATM Machine, draw an activity diagram that shows the data processing involved when you cash from the ATM machine? **[10 Marks]**

8. Discuss each of the following types of software maintenance:

   a. Corrective **[4 Marks]**
   b. Adaptive **[4 Marks]**
   c. Perfective **[4 Marks]**
   d. Preventive **[4 Marks]**

**END OF QUESTION PAPER**